

Question

试证明 $x = \wp\left(\frac{w}{5}\right) + \wp\left(\frac{2w}{5}\right)$, 其中 $w = 2m\omega + 2m'\omega'$, 是六次方程

$$x^6 - 5g_2x^4 - 40g_3x^3 - 5g_2^2x^2 - 8g_2g_3x - 5g_3^2 = 0$$

的根, 并把六个根都表示出来.¹

证明. 设 $\Lambda = \{2m\omega + 2m'\omega' \mid m, m' \in \mathbb{Z}\}$ 为周期格. 考虑 $z = \frac{w}{5}$, 其中 $w \in \Lambda$, 即 $5z \in \Lambda$. 令 $p = \wp(z)$, $q = \wp(2z)$, 则 $x = p + q$.

由倍角公式:

$$q = \frac{p^4 + \frac{g_2}{2}p^2 + 2g_3p + \frac{g_2^2}{16}}{4p^3 - g_2p - g_3}$$

去分母整理得

$$p^4 - 4p^3q + \frac{g_2}{2}p^2 + g_2pq + 2g_3p + g_3q + \frac{g_2^2}{16} = 0 \quad (1)$$

由 $5z \in \Lambda$ 得 $4z \equiv -z \pmod{\Lambda}$, 故 $\wp(4z) = \wp(-z) = \wp(z) = p$. 又由倍角公式, $\wp(4z) = \wp(2 \cdot 2z) = \frac{q^4 + \frac{g_2}{2}q^2 + 2g_3q + \frac{g_2^2}{16}}{4q^3 - g_2q - g_3}$, 因此有

$$q^4 - 4q^3p + \frac{g_2}{2}q^2 + g_2pq + 2g_3q + g_3p + \frac{g_2^2}{16} = 0 \quad (2)$$

将 (1) 与 (2) 相加, 得

$$p^4 + q^4 - 4pq(p^2 + q^2) + \frac{g_2}{2}(p^2 + q^2) + 2g_2pq + 3g_3(p + q) + \frac{g_2^2}{8} = 0$$

因 $x = p + q$, 设 $y = pq$, 则 $p^2 + q^2 = x^2 - 2y$, $p^4 + q^4 = x^4 - 4x^2y + 2y^2$, 代入得

$$x^4 - 4x^2y + 2y^2 - 4y(x^2 - 2y) + \frac{g_2}{2}(x^2 - 2y) + 2g_2y + 3g_3x + \frac{g_2^2}{8} = 0$$

化简为

$$x^4 - 8x^2y + 10y^2 + \frac{g_2}{2}x^2 + g_2y + 3g_3x + \frac{g_2^2}{8} = 0 \quad (A)$$

将 (1) 与 (2) 相减, 得

$$(p^4 - q^4) - 4pq(p^2 - q^2) + \frac{g_2}{2}(p^2 - q^2) + g_3(p - q) = 0$$

即

$$(p - q) \left(p^3 + q^3 - 3pq(p + q) + \frac{g_2}{2}(p + q) + g_3 \right) = 0$$

一般 $p \neq q$, 故约去 $p - q$ 得:

$$(p^3 + q^3) - 3pq(p + q) + \frac{g_2}{2}x + g_3 = 0$$

代入 $p^3 + q^3 = x^3 - 3xy$,

$$x^3 - 6xy + \frac{g_2}{2}x + g_3 = 0 \quad (B)$$

由 (B) 解得

$$y = \frac{g_2x + 2g_3 + 2x^3}{12x}$$

代入 (A) 即得

$$x^6 - 5g_2x^4 - 40g_3x^3 - 5g_2^2x^2 - 8g_2g_3x - 5g_3^2 = 0$$

因此, $x = \wp(z) + \wp(2z)$ 满足该六次方程.

¹该问题出自王竹溪《特殊函数概论》第八章外氏椭圆函数. 更多讨论请看: <https://chaoli.club/index.php/11858/0/>

由对称性, 只需取 (m, m') 模 5 下非零且互不等价的六个代表元. 例如:

$$x_{(1,0)} = \wp\left(\frac{2\omega}{5}\right) + \wp\left(\frac{4\omega}{5}\right),$$

$$x_{(0,1)} = \wp\left(\frac{2\omega'}{5}\right) + \wp\left(\frac{4\omega'}{5}\right),$$

$$x_{(1,1)} = \wp\left(\frac{2\omega + 2\omega'}{5}\right) + \wp\left(\frac{4\omega + 4\omega'}{5}\right),$$

$$x_{(1,2)} = \wp\left(\frac{2\omega + 4\omega'}{5}\right) + \wp\left(\frac{4\omega + 8\omega'}{5}\right),$$

$$x_{(1,3)} = \wp\left(\frac{2\omega + 6\omega'}{5}\right) + \wp\left(\frac{4\omega + 12\omega'}{5}\right),$$

$$x_{(1,4)} = \wp\left(\frac{2\omega + 8\omega'}{5}\right) + \wp\left(\frac{4\omega + 16\omega'}{5}\right),$$

其中 (m, m') 对应 $w = 2m\omega + 2m'\omega'$. 这六个数即为六次方程的全部根.

□

Mathematica 数值验证:

```
ClearAll["Global`*"]
g2 = 2.; g3 = 3.; (*不加小数点, 会报错*)
NSolve[x^6 - 5*g2*x^4 - 40*g3*x^3 - 5*g2^2*x^2 - 8*g2*g3*x -
  5*g3^2 == 0, x]
{\[Omega], \[Omega]p} = WeierstrassHalfPeriods[{g2, g3}];
wp[z_] := WeierstrassP[z, {g2, g3}];
wp[2*\[Omega]/5] + wp[4*\[Omega]/5] // N(*x1*)
wp[2*\[Omega]p/5] + wp[4*\[Omega]p/5] // N(*x2*)
wp[2 (\[Omega] + \[Omega]p)/5] +
  wp[4 (\[Omega] + \[Omega]p)/5] // N(*x3*)
wp[(2*\[Omega] + 4*\[Omega]p)/5] +
  wp[(4*\[Omega] + 8*\[Omega]p)/5] // N(*x4*)
wp[(2*\[Omega] + 6*\[Omega]p)/5] +
  wp[(4*\[Omega] + 12*\[Omega]p)/5] // N(*x5*)
wp[(2*\[Omega] + 8*\[Omega]p)/5] +
  wp[(4*\[Omega] + 16*\[Omega]p)/5] // N(*x6*)
```

输出:

```
{ {x -> -2.76545 - 3.65242 I}, {x -> -2.76545 + 3.65242 I}, {x -> -0.591168}, {x ->
  0.227847 - 0.766911 I}, {x -> 0.227847 + 0.766911 I}, {x -> 5.66638} }
5.66638 + 0. I
-2.76545 - 3.65242 I
0.227847 - 0.766911 I
0.227847 + 0.766911 I
-0.591168 + 2.2849*10^-16 I
-2.76545 + 3.65242 I
```